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Sleep architecture is altered with travel and matches in professional rugby union players

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ABSTRACT

Field-based athlete research has primarily focused on sleep-wake patterns, but technological advancements now enable detailed assessment of sleep architecture. This study compared sleep quantity, quality, and architecture during home (no travel; HOME) and away (eastbound travel across three time zones; AWAY) matches in professional male rugby union. The sleep of twenty athletes was assessed using home-based polysomnography over three nights per condition: two nights preceding the match (MD-2), match night (MD), and the following night after (MD + 1). Sleep duration, efficiency, onset/offset, latency, wake after sleep onset, awakenings, and sleep stages (as light, deep and rapid eye movement [REM] sleep, evaluated as proportion [%] and time [min]) were monitored. Compared to HOME, AWAY increased sleep onset latency (+12 ± 23 min; $p < 0.01$) and reduced light sleep (−6.2 ± 7.1%; $p < 0.01$). Match days (MD) incurred delayed bedtime (+71 ± 135 min; $p < 0.001$), reduced sleep duration (−94 ± 120 min; $p < 0.05$), and altered sleep architecture (deep: +9.8 ± 10.0%; REM: −6.9 ± 8.3%; both $p < 0.05$) compared to MD-2. On MD + 1, sleep duration (+96 ± 147 min; $p < 0.01$) and light and REM sleep time rebounded (+52 ± 44 min and + 39 ± 52 min, respectively; both $p < 0.05$). Travel and matches altered sleep architecture in professional rugby. Training and flight schedules should ensure sufficient sleep opportunity after travel and matchday to optimise recovery.

KEYWORDS
Sleep architecture; travel; rugby; team-sport; athletes; polysomnography

Introduction

The advancement of wearable sleep technology has shifted field-based sleep assessments from subjective to objective data collection (Halson, 2019; Kölling et al., 2019). This transition has highlighted the susceptibility of elite team-sport athletes to irregular sleep patterns (Fullagar et al., 2015; Halson, 2019; Kölling et al., 2019). Consequently, there has been a surge in field-based research over the past decade aimed at better understanding sleep to optimise athlete performance, recovery and welfare. Ongoing research continues to uncover the various mechanisms influencing sleep in athletes of professional teams.

Individual factors (i.e. age, chronotype, diet, and stress) contribute to sleep variability among players within a team (Nedelec et al., 2018; Savis, 1994). However, different training and competition phases also impact players' restorative sleep (Caia et al., 2017). Athletes face numerous external stressors, notably matches and travel, which can disrupt their sleep patterns (Caia et al., 2017; Fullagar et al., 2015). Match-related stressors and athlete behaviours including caffeine ingestion, muscular soreness, crowd stimulation, light arousal, and post-match commitments (i.e. press conferences) can compromise sleep quantity and quality, especially with evening kick-offs (Dunican et al., 2018; Juliff et al., 2015). Further, travel exposes players to stressors such as travel fatigue, jet lag, and unfamiliar sleeping environments – all of which can perturb sleep on nights away from home, ultimately disrupting match preparation (Janse van Rensburg et al., 2021). In professional rugby, athletes experience sleep restriction every weekend throughout a multiweek season (e.g. 15–27 weeks in Super Rugby Pacific, National Rugby League, United Rugby Championship). Accumulated sleep debt over consecutive weeks could potentially increase the risk of performance deterioration, injury, and deleterious health consequences (Charest & Grandner, 2022; Janse van Rensburg et al., 2021). Therefore, consistently achieving sufficient restorative sleep is crucial for both individual performance and overall team success.

Field-based research on objective sleep assessment in professional athletes during a competitive season has primarily relied on dichotomous sleep-wake analyses from wrist activity monitors to distinguish between sleep and wake states (Charest & Grandner, 2022; Halson, 2019). While studies indicate that matches and travel can significantly reduce sleep quantity in team-sport athletes, findings on sleep quality during these nights are inconsistent (Erlacher et al., 2009;

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Richmond et al., 2004; Sargent et al., 2022). For example, female volleyball players subjectively report poorer sleep quality during away than home matches (Erlacher et al., 2009) whereas male Australian rules footballers report the opposite (Richmond et al., 2004). Furthermore, an objective assessment in a similar cohort (Australian rules football) found no effect of match location on sleep quality across all nights surrounding matches (Sargent et al., 2022). Surrogate markers such as sleep efficiency, as observed by (Sargent et al., 2022) derived from dichotomous sleep-wake analyses, are currently the only available method for assessing sleep quality, apart from subjective evaluations upon waking (Halson, 2019; Savis, 1994). Accordingly, a gap exists in the literature regarding the detailed structural sleep organisation (i.e. sleep architecture) of athletes during a competitive season.

Sleep architecture refers to the organisation of sleep stages, including wakefulness, across the sleep period. The sleep stages are delineated into non-rapid eye movement (NREM) and rapid eye movement (REM) sleep, with NREM further categorised into light (N1 and N2) and deep (N3) sleep (Savis, 1994). Each sleep stage serves distinct physiological purposes: deep sleep (slow-wave sleep) is critical for physical and physiological restoration, while REM sleep supports mental and cognitive restoration (Savis, 1994). Therefore, understanding the sleep architecture of team-sport athletes during a season could help address issues related to sleep debt and performance deterioration following matches and travel.

The gold standard for assessing sleep architecture is via polysomnography (PSG), which utilises electroencephalography along with electrooculography and electromyography to distinguish between different sleep or wake stages (Savis, 1994). Despite this, the use of PSG in field-based research on athletes is constrained due to the need for laboratory settings, sleep technicians, and specialised equipment. Consequently, PSG is primarily employed as a one-time screening tool, conducted under controlled conditions to detect sleep disorders. Recent advancements in wireless home-based PSG devices enable researchers to assess sleep architecture in professional athletes across multiple nights in a home environment. This approach can enhance our understanding of sleep perturbations experienced by athletes during a competitive season (Driller et al., 2023; Miller et al., 2022). To date, only three studies have assessed in-season sleep architecture on team-sport athletes, including rugby union (Aloulou et al., 2020) soccer (Sanders et al., 2004) basketball, hockey and volleyball (Vitale et al., 2022). Two studies reported notable disruptions to sleep duration (Sanders et al., 2004) and delayed bedtime (Vitale et al., 2022) after a match, yet found no differences in sleep architecture both after either a match or across all surrounding nights (Sanders et al., 2004; Vitale et al., 2022). Interestingly, a study on youth rugby union athletes reported that match demands significantly altered the proportion of deep and light sleep after a match (Aloulou et al., 2020). Nonetheless, research on field-based sleep architecture in elite athletes remains sparse, highlighting the need for further investigation into travel- and match-related sleep disturbances, considering the importance of sleep on athletic performance (Charest & Grandner, 2022). Notably, no studies to date have specifically explored the sleep architecture of professional rugby union players during a competitive season.

Therefore, this field-based study aimed to compare the effects of a home and an away match fixture on sleep quantity, quality, and architecture in professional male rugby union players. We hypothesised that (1) short-haul eastbound travel would delay sleep onset and thereby reduce sleep duration when playing away; (2) matches would delay bedtime, increase sleep onset latency, and reduce total sleep time, regardless of location; (3) earlier wake time due to the return travel schedules after an away match would reduce REM sleep; and (4) after playing away, sleep duration (and time spent in all sleep stages) would significantly rebound the next night upon returning home.

Methods

Participants

Twenty-six male professional rugby union players, contracted for the Super Rugby season, were recruited for this study. All players regularly completed four field-based training sessions (rugby; 6 h) and three to four gym sessions (4.5 to 4.75 h) weekly. The study's purpose and methodology were explained to participants before informed consent was obtained. Ethics approval was provided by the host institution (2022/ET000700).

Experimental design

A prospective observational study was conducted during the first two rounds of the 2023 Super Rugby Pacific season (February to March 2023) to compare sleep (quantity, quality, and architecture) across two conditions: a HOME *vs*. an AWAY match fixture. Sleep was assessed over three nights for each condition: the two nights preceding the match (MD-2), the night of the match (MD), and the second night following matchday (MD + 1). Eight days separated the two matches, with HOME preceding AWAY. Randomisation was not possible due to field-testing logistics and the team fixture. Training schedule and match preparation were replicated across the two conditions, except for one gym session, which was omitted from the AWAY condition due to travel. A familiarisation session involved instructing participants on how to wear the sleep monitor; however, they did not undergo a familiarisation night of data collection.

Travel for AWAY involved a direct, short-haul (3.6 h), eastward flight across three time zones on MD-2 (departing at 13:10 [GMT +8] and arriving at 19:45 [GMT +11]). Dinner was served immediately upon arrival at the hotel and players went to bed at their discretion. Following the match, players returned westward on a flight the morning of MD + 1 (departure at 09:35 [GMT +11]; arrival at 11:15 [GMT +8]). All travel was in economy class. Overall, one-way travel duration was ~4.9 h including bus transfers (training headquarters to airport [30 min] and airport to hotel [45 min]). Standard team travel management strategies (compression garments, hydration, and nutritional aids) were employed by all players per team protocol.

Sleep

Players wore a wireless home-based PSG device (Somfit, Compumedics, Melbourne, Australia) on their forehead during designated sleep assessment nights. The Somfit has been validated against PSG to assess sleep architecture (Miller et al., 2022). Players were instructed to wear the device immediately before bed and remove it upon waking (once out of bed). The data was uploaded to the proprietary viewing platform (Nexus360; Compumedics, Melbourne, Australia) and automatically analysed. Sleep staging was then manually scored in 30-s epochs by a single experienced sleep scientist into N1, N2, N3 and REM sleep (or wake), according to recommended criteria (Berry Richard et al., 2017).

Sleep parameters were categorised into three subgroups (Driller et al., 2023; Halson, 2019):

- *Sleep quantity*: total sleep time.
- *Sleep quality*: sleep efficiency (total sleep time as a percentage of time in bed).
- *Sleep architecture*: sleep stages were categorised into light (N1 and N2), deep (N3) and REM sleep (and presented as both proportion [%] and duration [min] per sleep stage), sleep onset time, sleep offset time, sleep onset latency, REM onset latency, wake after sleep onset (WASO) and the number of awakenings per night.

Statistical analysis

A *priori* power analysis based on previous literature22 ($\alpha = 0.05$, statistical power = 90%, and effect size [η_p2; derived from total sleep time] = 0.68) determined a minimum sample of eight participants. Data are presented as mean±standard deviation (SD). All statistical analyses were undertaken on RStudio (Version 4.3). Linear mixed-effects regressions (lmer) were used to analyse the effect of the HOME *vs*. AWAY on sleep variables. The model incorporated fixed covariates effect of 'condition' and 'day' while accounting for individual differences (*player* as a random intercept) using the linear mixed-effects models with 'Eigen' and S4 (lme4) package. Post-hoc pairwise comparisons with *Tukey*-adjusted *p*-values were performed using the estimated marginal means (emmeans) package. All missing sleep data were coded as 'NA', with no imputation. Statistical significance was set at $p < 0.05$. Effect sizes were reported as partial eta squared (η_p^2; small effect ≤ 0.01, *moderate* effect = 0.06; *large* effect ≥ 0.14) (Cohen, 1977).

Results

Out of the twenty-six participants that were recruited, six were excluded as they did not record sleep. Twenty male professional rugby union players (age: 27.5 ± 4.1 years; height: 185.4 ± 7.1 cm; weight: 99.6 ± 12.3 kg; positions: 9 forwards and 11 backs) participated in this study, each recording at least one valid night of sleep. Of the planned 120 sleep assessment nights (3 nights x 2 conditions x 20 participants), data were successfully collected for 69 nights (38 observations in HOME [13 ± 2 per night]; 31 observations in AWAY [10 ± 1 per night]). Non-adherence to experimental protocol resulted in missing nights.

Sleep quantity

Total sleep time was significantly influenced by day ($p < 0.01$; $\eta_p 2 = 0.15$; Table 1), regardless of condition. Sleep duration on MD was shorter than MD-2 (−94 ± 120 min; p=0.016) and MD + 1 (−96 ± 147 min; p=0.003).

Sleep quality

Sleep efficiency remained unaffected by day ($p = 0.700$) and condition ($p = 0.495$; Table 1).

Sleep architecture

Significant effects were found for the proportion of time spent in light (between conditions), deep (across days), and REM sleep (both days and conditions; Figure 1). Light sleep was influenced by condition ($p < 0.01$; $\eta_p 2 = 0.01$), with reductions when AWAY (−6.2 ± 7.1%; p=0.002) compared to HOME. Deep sleep revealed effects of both day ($p < 0.05$; $\eta_p 2 = 0.13$) and condition ($p < 0.05$; $\eta_p 2 = 0.03$); however, post-hoc analyses only identified differences between days, with a greater proportion on MD (+9.8 ± 10.0%; p=0.014) compared to MD-2. REM sleep was influenced by day ($p < 0.05$; $\eta_p 2 = 0.09$), with reductions on MD (−6.9 ± 8.3%; p=0.026) compared to MD-2.

Significant effects were found for the total time spent in light (across days), deep (both conditions and days), and REM sleep (across days; Figure 2). Light sleep differed across days ($p < 0.01$; $\eta_p 2 = 0.15$), with less time on MD compared to MD-2 (−69 ± 82 min; $p = 0.009$) and MD + 1 (−52 ± 94 min; $p = 0.022$). REM sleep followed a similar pattern ($p < 0.001$; $\eta_p 2 = 0.18$), with reduced time on MD compared to MD-2 (−42 ± 42 min; $p = 0.002$) and MD + 1 (−39 ± 52 min; $p < 0.001$). Deep sleep was affected by both day ($p < 0.05$; $\eta_p 2 = 0.01$) and condition ($p < 0.05$; $\eta_p 2 = 0.01$), with greater time on MD + 1 (+15 ± 23 min; $p = 0.042$), compared to MD-2, and when AWAY (+15 ± 24 min; $p = 0.044$) compared to HOME.

Time of sleep onset was influenced by day ($p < 0.001$; $\eta_p 2 = 0.31$; Table 1) but not condition, with players falling asleep later on MD compared to all other nights (both $p < 0.001$). However, wake times were not affected by day or condition (day: $p = 0.948$; condition: $p = 0.392$; Table 1).

Sleep onset latency differed between days ($p < 0.05$; $\eta_p 2 = 0.12$) and condition ($p < 0.01$; $\eta_p^2 = 0.14$), with players taking longer to fall asleep when AWAY (+12 ± 23 min; $p = 0.003$) compared to HOME, and on MD (+11 ± 27 min; $p = 0.035$) compared to MD + 1. Latency into REM sleep, as well as the number of awakenings and WASO remained unaffected (all $p > 0.05$; Table 1).

Discussion

This study is the first to assess sleep architecture in-season in professional rugby union players using home-based PSG. We reveal novel findings that sleep architecture is altered on nights following home and away match fixtures. When AWAY, players

Table 1. Descriptive statistics (mean±sd) of sleep quantity, quality, and architecture of professional male rugby union players across HOME and AWAY match fixtures.

Condition	Variables	MD-2	MD	MD +1	ANOVA *p*-value (ηp^2)	
Sleep quantity					Day	Condition
Home	Total sleep time, mins	412 ± 67†	306 ± 48$^{\dagger ¢}$	369 ± 164$^{¢}$	*p* <0.01	0.504
Away		343 ± 104	297 ± 89	455 ± 37	(0.153)	(0.002)
Sleep quality						
Home	Sleep efficiency, %	88.0 ± 9.1	88.8 ± 6.1	83.9 ± 21.9	0.699	0.495
Away		86.3 ± 11.3	78.1 ± 19.3	90.3 ± 4.0	(0.015)	(0.006)
Sleep architecture						
Home	Light sleep, %	67.3 ± 5.7*	60.9 ± 7.3	62.1 ± 14.8	0.263	*p* <0.01
Away		59.0 ± 7.0*	58.6 ± 6.3	57.7 ± 7.4	(0.035)	(0.079)
Home	Deep sleep, %	9.5 ± 6.8†	20.1 ± 7.0†	18.1 ± 14.1	*p* <0.05	*p* <0.05
Away		17.9 ± 6.8	23.5 ± 9.1	14.2 ± 5.8	(0.131)	(0.030)
Home	REM sleep, %	23.5 ± 5.5†	19.3 ± 7.8†	20.6 ± 9.0	*p* <0.05	0.381
Away		23.4 ± 6.7	18.1 ± 7.5	28.2 ± 4.8	(0.089)	(0.011)
Home	Light sleep, mins	278 ± 54†	187 ± 40$^{\dagger ¢}$	227 ± 107$^{¢}$	*p* <0.01	0.063
Away		201 ± 63	174 ± 51	261 ± 28	(0.151)	(0.032)
Home	Deep sleep, mins	36 ± 21*†	60 ± 16†	58 ± 33	*p* <0.05	*p* <0.05
Away		58 ± 20*	65 ± 22	65 ± 28	(0.087)	(0.056)
Home	REM sleep, mins	98 ± 30†	60 ± 25$^{\dagger ¢}$	85 ± 54$^{¢}$	*p* <0.001	0.645
Away		85 ± 39	58 ± 32	129 ± 30	(0.184)	(0.003)
Home	Sleep onset, hh:mm	22:17 ± 00:49†	01:50 ± 00:58$^{\dagger ¢}$	23:27 ± 01:26$^{¢}$	*p* <0.001	0.939
Away		00:45 ± 00:29	00:18 ± 00:58	21:59 ± 00:47	(0.311)	(0.002)
Home	Sleep offset, hh:mm	05:54 ± 00:34	07:13 ± 01:15	06:40 ± 01:34	0.948	0.392
Away		07:07 ± 01:54	05:42 ± 01:29	05:58 ± 00:33	(<0.001)	(0.008)
Home	Sleep onset latency, mins	11 ± 11*	15 ± 15$^{¢}$	10 ± 8$^{¢}$	*p* <0.05	*p* <0.01
Away		19 ± 10*	41 ± 33	18 ± 19	(0.120)	(0.140)
Home	REM latency, mins	91 ± 35	138 ± 47	112 ± 84	0.061	0.136
Away		94 ± 37	116 ± 37	79 ± 37	(0.078)	(0.017)
Home	Wake after sleep onset, mins	38 ± 26	21 ± 10	36 ± 33	0.500	0.932
Away		30 ± 35	33 ± 21	35 ± 18	(0.020)	(<0.001)
Home	Awakenings, n	21 ± 6	20 ± 7	19 ± 11	0.458	0.436
Away		17 ± 8	16 ± 8	24 ± 7	(0.030)	(0.009)

†*Denotes a significant difference between MD-2 and MD, regardless of condition (p < 0.05).*
$^{¢}$*Denotes a significant difference between MD and MD + 1, regardless of condition (p < 0.05).*
**Denotes a significant difference between HOME and AWAY, regardless of day (p < 0.05).*

experienced reductions in light sleep (proportion) and increased deep sleep (time), alongside difficulty falling asleep. After a match, regardless of location, significant sleep curtailment was observed, unveiling reductions in light and REM sleep (time) and greater deep sleep (time). This altered the overall sleep architecture (lower proportion of REM; greater proportion of deep sleep) after a match when compared to MD-2. Following this, sleep quantity, as well as light and REM sleep (time) rebounded significantly on MD + 1, particularly following return travel from away matches. Sleep restriction induced by travel and matches confirmed our initial hypothesis, highlighting the impact on sleep disturbances in professional rugby union players during a competitive season. Our findings emphasise the importance of enabling sufficient sleep opportunities following travel and matchday in professional rugby.

Reductions to light sleep during an AWAY fixture

When AWAY, players experienced reduced light sleep (proportion), increased deep sleep (time), and greater difficulty falling asleep compared to HOME. Given that light sleep is predominantly made up of N2 sleep (as N1 represents ~ 5% and N2 represents ~ 45–55% of total sleep time), and that N1 sleep is necessary for the transition from wakefulness to sleep (Driller et al., 2023; Savis, 1994) the reduction in light sleep proportion observed in our study (−6.2 ± 7.1%) likely stems from N2 sleep restriction. This is similar to findings from a laboratory study assessing non-athletes enduring a simulated 5-h phase advance (Petit et al., 2014). While the exact stimulus for this change in sleep architecture remains unclear, previous studies suggest that when total sleep time is reduced, N2 sleep is often affected (Shapiro, 1982; Van Dongen et al., 2011). In our study, although total sleep time did not significantly differ between conditions, compared to HOME MD-2 (which did not require travel), sleep reduced by 50 ± 51 min on AWAY MD-2 (after travel), exceeding the clinically meaningful change threshold of 20 min[24]. Future investigations should address reasons for reduced light sleep (proportion) on nights spent away from home.

The restorative role of N2 sleep has been debated (Savis, 1994) but recent research indicates it may influence subjective sleep satisfaction, albeit in non-athletic cohorts (Lok et al., 2022). This could explain the frequent sleep dissatisfaction reported by athletes following short- (Richmond et al., 2004) and long-haul travel (Fowler et al., 2015). Importantly, previous studies indicate that greater travel demands (i.e. longer durations and/or greater circadian shifts) exacerbate effects on sleep (Fowler et al., 2015). Given that this study involved a 3-h phase advance affecting sleep architecture, future studies should explore the impact of more extensive and frequent travel (a common requirement in professional rugby) (Dunican et al., 2018) to enhance travel recovery and match preparation.

Our study confirms the challenges of eastbound transmeridian travel and the potential unfamiliarity of hotel sleep (Janse van Rensburg et al., 2021; Petit et al., 2014) as

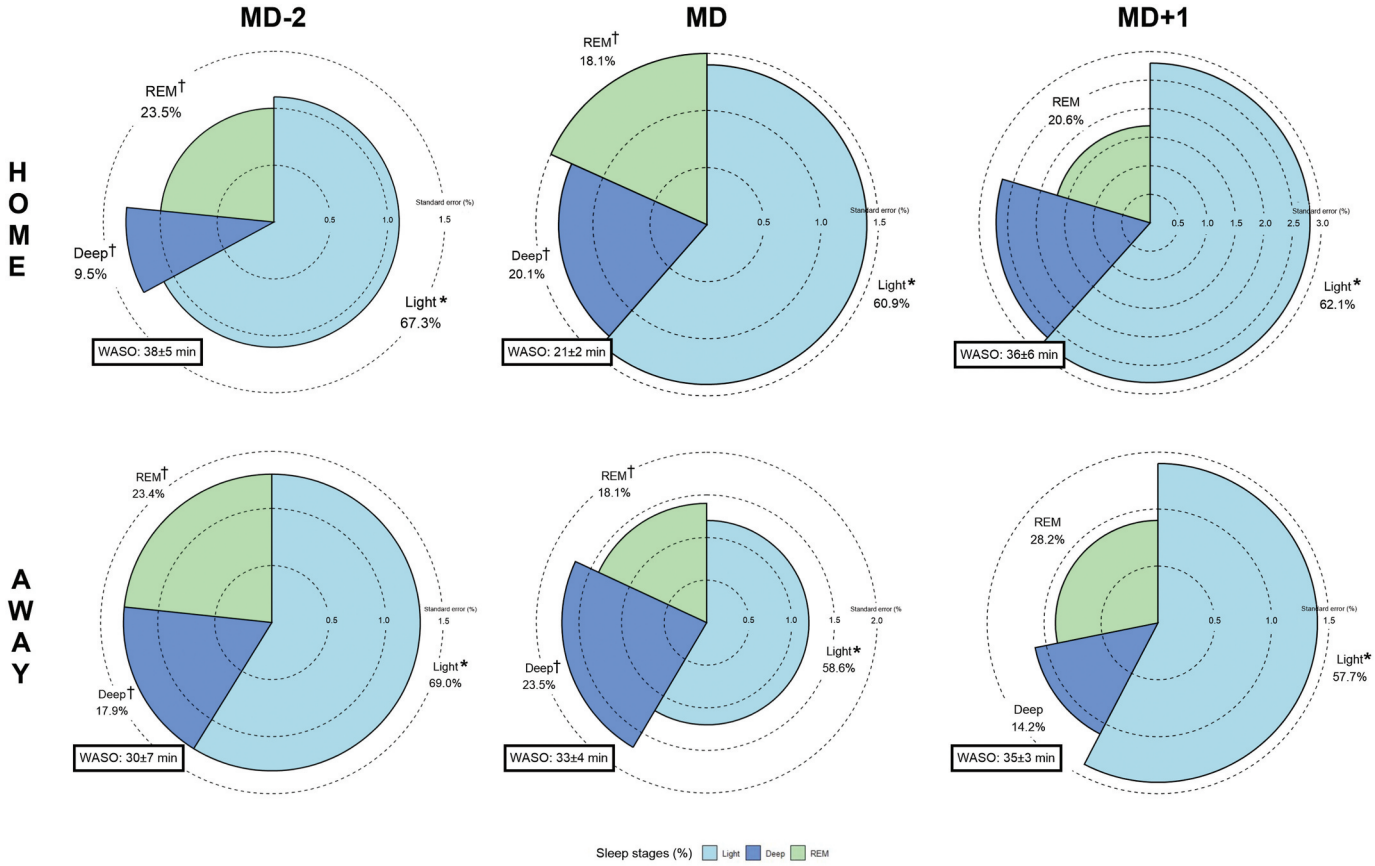

Figure 1. Proportions of the sleep period spent in each sleep stage (light, deep and REM), as well as wake after sleep onset (WASO) of a professional male rugby union team ($n = 20$) across a HOME and AWAY match fixture. The segments of the coxcomb chart represent the mean values of each sleep stage, with the depth of each segment representing the standard error of the data. WASO values are presented in text as mean and standard error. * *denotes a significant difference between HOME and AWAY, regardless of day ($p < 0.05$),* † *denotes a significant difference between MD-2 and MD, regardless of condition ($p < 0.05$).*

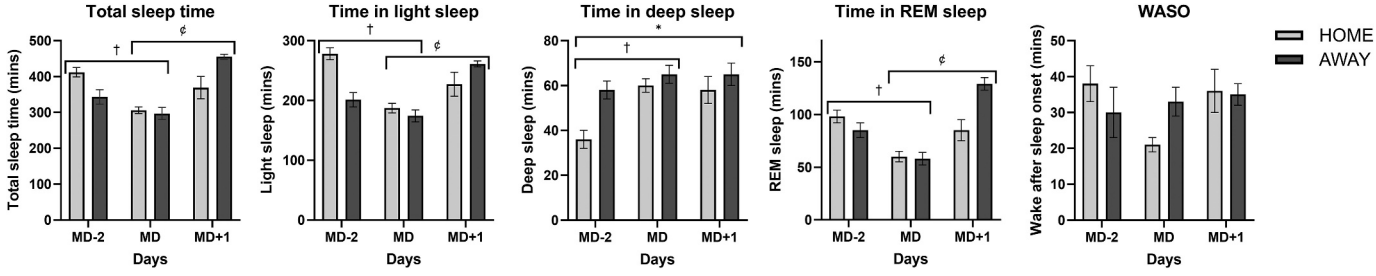

Figure 2. Time spent in each sleep stage, as well as, the total sleep duration and wake after sleep onset (WASO) of a professional male rugby union team ($n = 20$) across a HOME and AWAY match fixture. The bar graph represents the mean values and the error bars represent the standard error of the data. * *denotes a significant difference between HOME and AWAY ($p < 0.05$),* † *denotes a significant difference between MD-2 and MD ($p < 0.05$),* ¢ *denotes a significant difference between MD and MD + 1 ($p < 0.05$).*

players took longer to fall asleep when AWAY (+12 ± 23 min; presenting clinically meaningful change (Sateia Michael et al., 2017) compared to HOME. Furthermore, while not statistically significant, players went to bed ~2.5 h later on AWAY MD-2 (upon arrival) compared to HOME. This aligns with the commonly reported delayed sleep propensity after eastbound travel, which requires a phase advance in the circadian rhythm (Janse van Rensburg et al., 2021). After further analysis to assess the effects of sleeping environments (coding AWAY MD + 1 as HOME due to return travel), hotel sleep significantly delayed sleep onset by + 1:26 ± 1:52 h (p=0.004) compared to HOME. Furthermore, hotel sleep significantly reduced total sleep duration (−87 ± 87 min; p=0.004; presenting clinically meaningful change (Sateia Michael et al., 2017) and altered sleep architecture (light: −4.3 ± 8.3% [p=0.045]; deep: +6.8 ± 7.4% [p=0.011]) in a similar manner. Despite a limited understanding of hotel *vs*. home sleep in athletic cohorts, research highlights the importance of creating good sleep environments (e.g. choice of pillows, ergonomic mattresses, cool temperature, minimal lighting/noise) on hotel sleep quality (Mao et al., 2018). Emphasising sleep environments (dark, quiet, cool) and practising good sleep hygiene in the new environment (i.e. wind-down routines, as well as avoiding electronic devices and

caffeine) may aid in reducing difficulties falling asleep in travelling players. Teams may also request rooms on specific floors or sides (e.g. west-facing) to minimise ambient noise and natural sunlight, further reducing sleep disturbances overnight. Furthermore, later training start times (e.g. an afternoon captain's run) may help preserve sleep at the new location.

Post-match sleep curtailment reduces light and REM sleep while preserving deep sleep

Regardless of condition, players experienced delays in sleep onset and significant reductions in overall sleep quantity the night after a match, compared to MD-2. This confirms the well-documented challenges of post-match sleep (Dunican et al., 2018; Vitale et al., 2022). This study also unveils the novel finding that sleep architecture is altered in professional team-sport athletes following a match, with reductions in light (time) and REM sleep (proportion and time), alongside increased deep sleep (proportion and time) compared to sleep on MD-2. Typically, deep sleep occurs primarily in the first half of the sleep period, whilst REM sleep occurs mostly in the latter half (Savis, 1994). Further, N2 sleep, which predominantly makes up light sleep, is interspersed during REM sleep, and therefore, also largely achieved in the latter half (Savis, 1994). As such, the combination of later bedtimes and early wake likely shortened overall sleep opportunity. Specifically, after an AWAY match, an early return travel schedule forced players to wake significantly earlier than HOME (05:42 ± 01:29 *vs*. 07:13 ± 01:15). The disrupted sleep-wake patterns, with players waking before achieving the recommended ~ 8–10 h sleep, likely led to the early termination of light and REM sleep restoration. The combination of shorter sleep opportunity, disrupting the latter half of the sleep period while leaving the first half undisrupted, likely explains the compromised light and REM sleep (time) observed in this study, alongside the preservation of deep sleep (time). This highlights the importance of ensuring athletes have adequate sleep opportunities. When AWAY, practitioners must carefully consider the travel schedules to preserve recovery after a match. Avoiding early morning return flights after an away match may help preserve post-match sleep.

While we hypothesised that reduced light sleep affects sleep satisfaction, REM sleep restriction can also negatively influence subsequent daytime function (Savis, 1994). Decreased REM sleep can deteriorate cognitive and psychological processes, often resulting in feelings of '*grogginess*' when waking from NREM sleep (especially deep sleep) rather than in REM[5]. Accordingly, REM sleep deficiency (i.e. diminished cognitive restoration) may impair attentiveness, information processing, and memory retention when learning new rugby content (Savis, 1994). Coaches may consider adapting their approach, content, or timing in relation to match reviews, typically held the morning after a match (especially with 5- or 6-day turnarounds), to accommodate potential cognitive impairments in players. Alternatively, they may delay start times to encourage adequate rest before match reviews. However, future studies should investigate the effects of altered sleep architecture on cognitive performance in professional athletes after both home and away matches.

An important consideration is the increased deep sleep (proportion and time) observed on MD compared to MD-2. This increase may partly reflect the body's greater need for physical recovery after a match, supporting the hypothesis that greater diurnal activity leads to greater deep sleep (Aloulou et al., 2020; Davenne, 2009). However, given the similar durations of deep sleep (time) across days (56 ± 26 min; with the exception of HOME MD-2), it is plausible that this significant increase is due to disturbed deep sleep on HOME MD-2 (36 ± 21 min). Therefore, we believe that matches, potentially driven by the heightened need for physical recovery, led to the preservation (rather than an increase) of deep sleep time. Overall, the greater deep sleep proportion on MD is likely a result of reduced light and REM sleep. Although the reason for reduced deep sleep on HOME MD-2 is unclear, factors such as daytime naps, which can reduce deep sleep in non-athletes (Dijk, 2010) or a potential first-night effect from using home-based PSG, may have played a role and warrants further investigation.

Rebounds in sleep architecture after night/s of sleep restriction

On MD + 1, regardless of condition, there were significant rebounds in total sleep time (+96 ± 147 min), as well as the time spent in light (+52 ± 94 min) and REM sleep (+39 ± 52 min) – all of which were reduced post-match. Although sleep quantity varies across the days (MD-2 *vs*. MD *vs*. MD + 1), the total sleep duration accumulated by the end of the fixture per condition was not different (~18 h across the three nights). While we did not assess sleep the night before the match due to logistical considerations, these findings highlight that sleep drive is determined by the amount of sleep deprivation from preceding nights. This likely explains the rebound effect, which is more exacerbated when players return from AWAY fixtures, incurring multiple nights of sleep restriction.

Rebound sleep at HOME on MD + 1 remained lower than on MD-2 (−36 ± 111 min; $p>0.05$; clinically meaningful change (Sateia Michael et al., 2017), suggesting that some players may not fully restore adequate sleep duration even on MD + 1, typically the night before the next training week begins. Poor sleep may signal inadequate recovery, which could signal residual fatigue. However, rebound sleep following large sleep debt or multiple nights of sleep restriction (e.g. returning home after being away), may mislead interpretations of sleep data, inappropriately suggesting players are adequately recovered. Therefore, high-performance staff need to consider the context when interpreting the sleep data of players. Future studies should include fatigue markers alongside sleep architecture to explore whether teams enter training (after a match/travel) with residual fatigue. If fatigue persists, it could hinder training, increase injury risk, and/or accumulate over the season (Charest & Grandner, 2022). This may be particularly relevant for teams with high travel demands and/or arduous fixture schedules across a season. Therefore, it becomes crucial to prioritise not only sleep but also recovery strategies (i.e. napping,

nutrition/hydration approaches, contrast therapy/cold water immersion, active recovery) to alleviate fatigue and expedite the overall recovery process.

Interindividual variability of sleep architecture amongst players within a team

Our sleep results showed significant variability across nights, confirming the differences in sleep patterns among players of a professional team (Caia et al., 2017; Dunican et al., 2018). This indicates that some players may be more susceptible to travel and/or match-related sleep disturbances. Variability in sleep and wake times, likely influenced by circadian preferences, can disturb the sleep quantity, quality, and architecture in certain individuals (Nedelec et al., 2018). For example, eastbound travel (as in this study) tends to disrupt the sleep-wake patterns of evening- rather than morning-type chronotypes (Janse van Rensburg et al., 2021). In contrast, when given opportunities to sleep-in on a training-free day (e.g. on MD + 1), the earlier wake preferences of morning-type chronotypes may limit their ability to extend their total sleep time, compared to evening-types that tend to naturally wake later (Janse van Rensburg et al., 2021). Consequently, our results suggest a large variance in wake times across players: i) after the first night of hotel sleep (AWAY MD-2), ii) after post-match sleep (HOME and AWAY), and iii) when given the opportunity to sleep-in (HOME MD + 1). As a result, high variability in certain sleep parameters is observed on these nights. For example, a large variance in sleep efficiency is presented on AWAY MD (78 ± 19%) and HOME MD + 1 (84 ± 22%). Factors such as diet and daytime commitments on training-free days (MD + 1) may further contribute to this variability. When delaying training start times, staff should communicate the rationale to players, encouraging them to extend their sleep opportunity and support recovery. This may also apply to training-free days.

Objective sleep monitoring can help staff identify individuals susceptible to travel- and/or match-induced sleep disturbances to provide personalised interventions (e.g. education, sleep hygiene, medication) and/or identify other factors (i.e. high training loads) that might increase the risk of sleep irregularity for the entire team. Variability of sleep highlights the importance of regularly monitoring sleep via objective measures during a season, or at least during each match fixture (with higher priority when playing away from home). Sleep data could enable staff to make informed decisions on training schedule adjustments (e.g. start times) or training modifications to mitigate any injury risk if sleep is significantly disturbed for certain individuals (Charest & Grandner, 2022). However, compliance with sleep assessments is necessary. Sleep monitoring on nights after matches and nights spent away from home should be prioritised.

Limitations

The limitations of this study must be acknowledged. Firstly, as a field-based study, we were unable to control for factors such as napping behaviour, caffeine or alcohol intake, electronic device use and parental commitments. Additionally, match schedules differed between HOME (19:00 local time) and AWAY (15:00 local time) games. While previous literature states that evening matches cause greater sleep restriction than daytime matches (Vitale et al., 2022) our study determined that even afternoon matches significantly delayed sleep onset. Furthermore, our conclusions may not be directly applicable to other periods of the season (i.e. differing schedules and/or match fixtures later in the season). Secondly, due to logistical considerations (i.e. psychological factors, disruption of preparatory routines) for competing athletes, we did not assess sleep the night before matches (MD-1) or conduct physical assessments. While such data could provide further insights, our primary objective was to examine sleep after travel and matches without burdening athletes' preparatory routines for competition with additional assessments. Finally, although the sleep device used provided valid sleep architecture data (Berry Richard et al., 2017; Miller et al., 2022) players reported sensor discomfort due to its forehead placement, leading to compliance issues, as also noted in Vitale et al (Vitale et al., 2022). Therefore, while the device may be suitable for short-term studies, its use in longitudinal research may be impractical. Teams may consider using validated wrist- or finger-worn sleep devices that infer sleep architecture to improve compliance (Driller et al., 2023; Miller et al., 2022).

Conclusion

Professional rugby union players experience alterations in sleep quantity, quality, and architecture on nights surrounding both home and away matches. Travelling can reduce light sleep (proportion) and increase the time to fall asleep. Matches, regardless of location, can significantly decrease total sleep time, light sleep (time), and REM sleep (proportion and time), whilst preserving deep sleep (proportion and time) in players. Teams should arrange in-season training and flight schedules to ensure sufficient sleep following travel and matches to optimise recovery. Competition factors, such as travel and matches, can pose distinct challenges to sleep architecture, with significant variability amongst individuals.

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